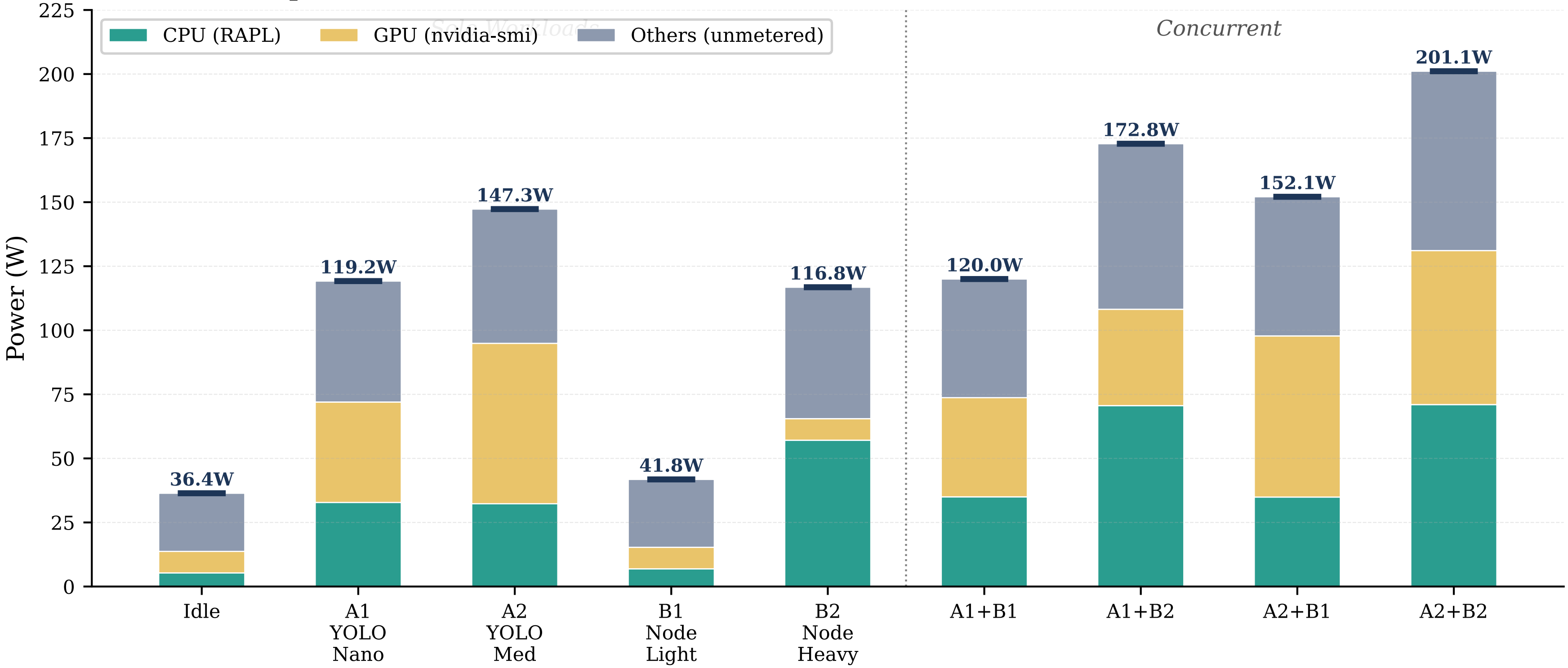
Phase 1.6 — Workload Power Breakdown (9 Scenarios)

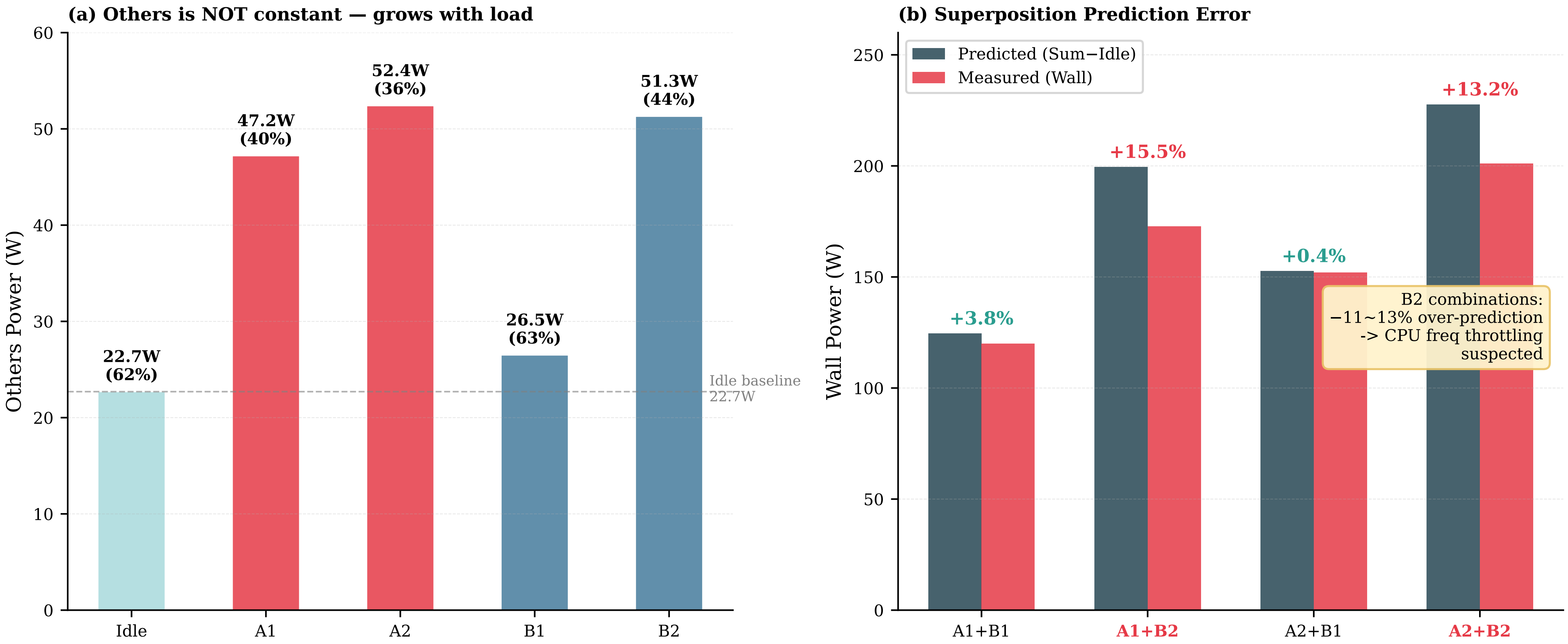
(a) Power Component Stacked Bar



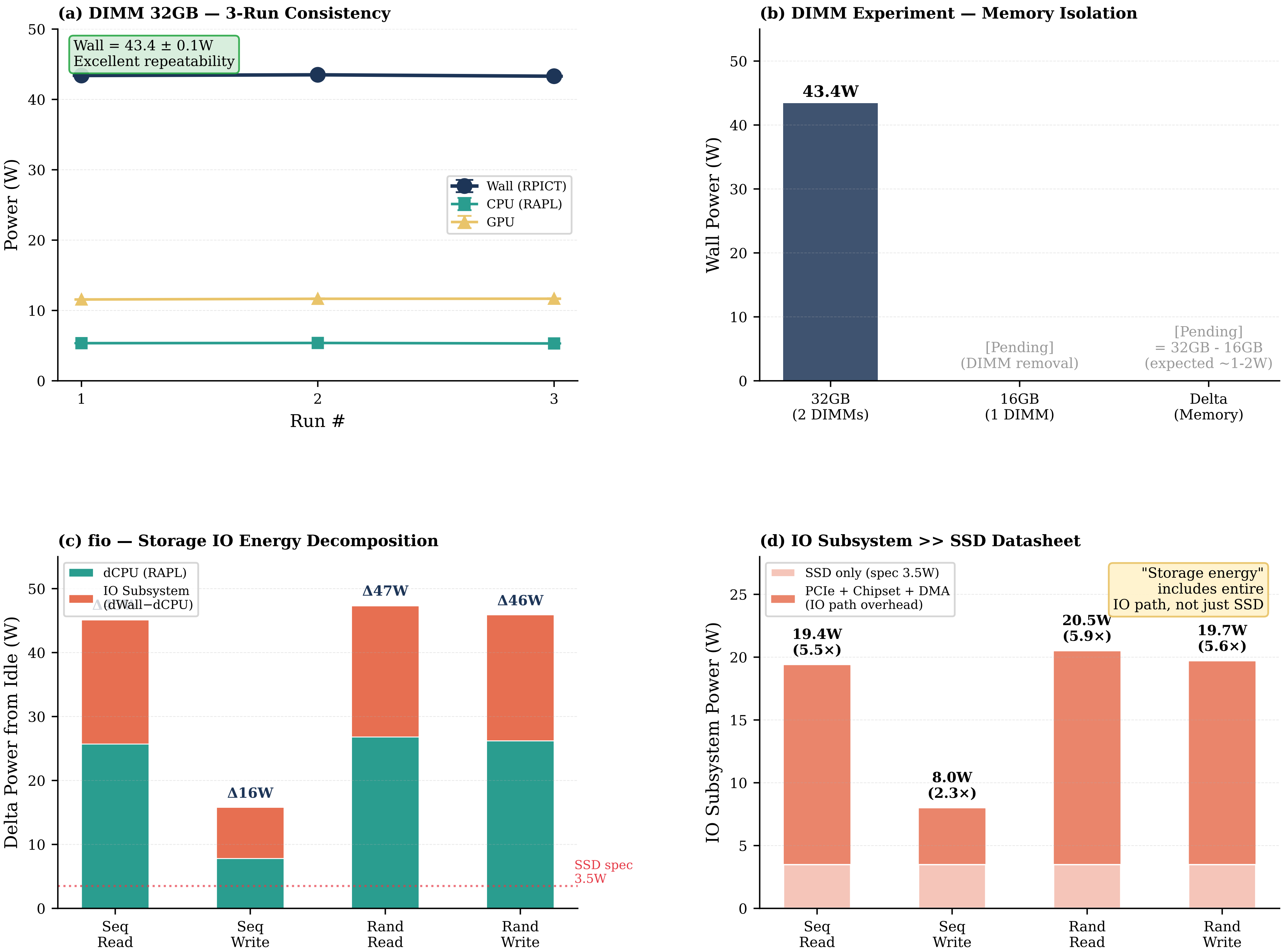
(b) Full Measurement Table (Phase 1.6)

	Idle	A1 YOLO Nano	A2 YOLO Med	B1 Node Light	B2 Node Heavy	A1+B1	A1+B2	A2+B1	A2+B2
Wall (W)	36.4	119.2	147.3	41.8	116.8	120.0	172.8	152.1	201.1
CPU (W)	5.3	32.8	32.3	6.9	57.1	35.0	70.6	34.9	71.0
GPU (W)	8.4	39.2	62.6	8.4	8.4	38.7	37.6	62.9	60.1
Others (W)	22.7	47.2	52.4	26.5	51.3	46.3	64.6	54.3	70.0
Others %	62.4%	39.6%	35.6%	63.4%	43.9%	38.6%	37.4%	35.7%	34.8%
dWall	—	+82.8	+110.9	+5.4	+80.4	+83.6	+136.4	+115.7	+164.7
dCPU	—	+27.5	+27.0	+1.6	+51.8	+29.7	+65.3	+29.6	+65.7
dGPU	—	+30.8	+54.2	+0.0	+0.0	+30.3	+29.2	+54.5	+51.7
dOthers	—	+24.5	+29.7	+3.8	+28.6	+23.6	+41.9	+31.6	+47.3

Phase 1.6 — "Others" Energy Analysis

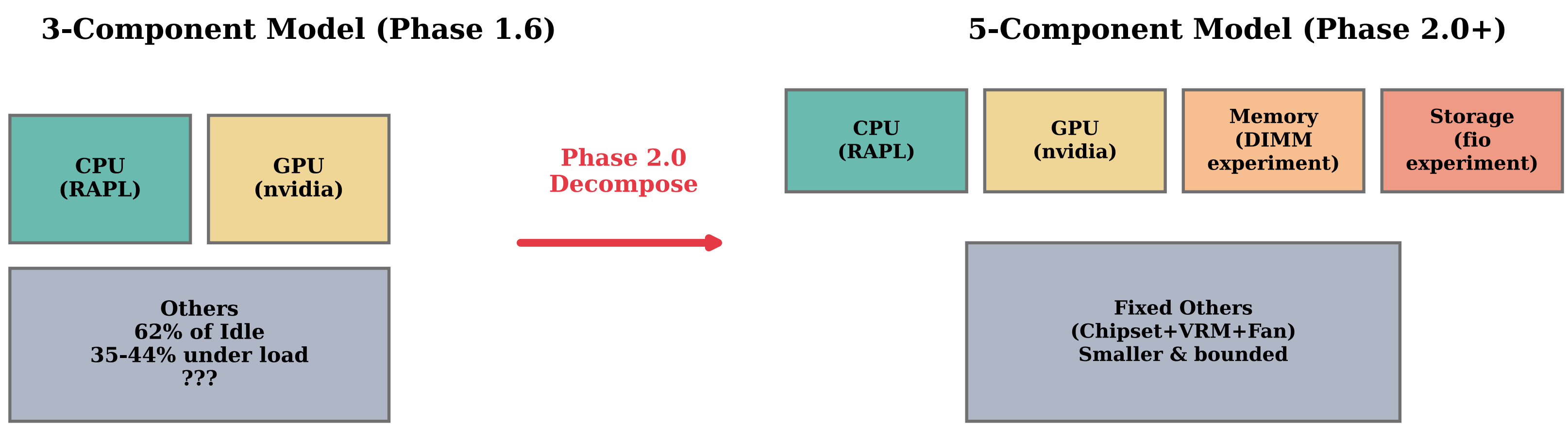


Phase 2.0 — Decomposing "Others" via Physical Measurement

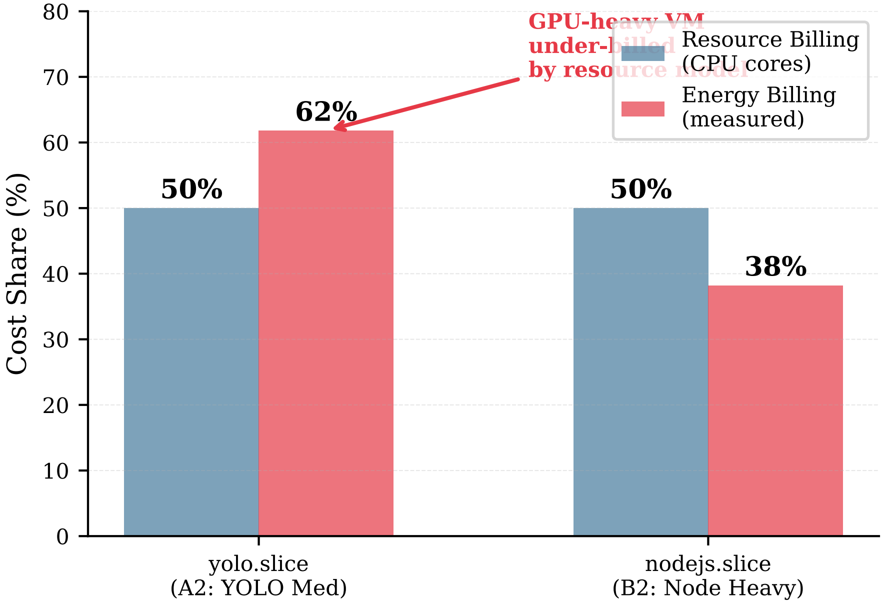


Integrated Summary — From 3-Component to 5-Component Model

(a) Attribution Model Evolution



(b) Resource Billing != Energy Billing (A2+B2 concurrent scenario)



(c) fio Storage Experiment Results

Phase	Wall (W)	dWall (W)	dCPU (W)	IO Sub. (W)	Throughpu
Idle	43.4	—	—	**—**	—
Seq Read	88.6	+45.1	+25.7	**19.4**	16.4 GB/s
Seq Write	59.2	+15.8	+7.8	**8.0**	2.2 GB/s
Rand Read	90.8	+47.3	+26.8	**20.5**	313K IOPS
Rand Write	89.3	+45.9	+26.2	**19.7**	291K IOPS